Statistics & Error Analysis

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ **Contents & learning path**

1. A measurement is a number with a doubt
2. Mean, median, spread and standard error
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1. A measurement is a number with a doubt

No astronomical measurement is exact. Statistics tells us how to summarize repeated readings; error analysis tells us how uncertainty moves through a calculation. A result should be written as $x \pm \sigma_x$ with units, and the meaning of σ_x should be clear: standard deviation of individual readings, standard error of the mean, or a combined uncertainty.

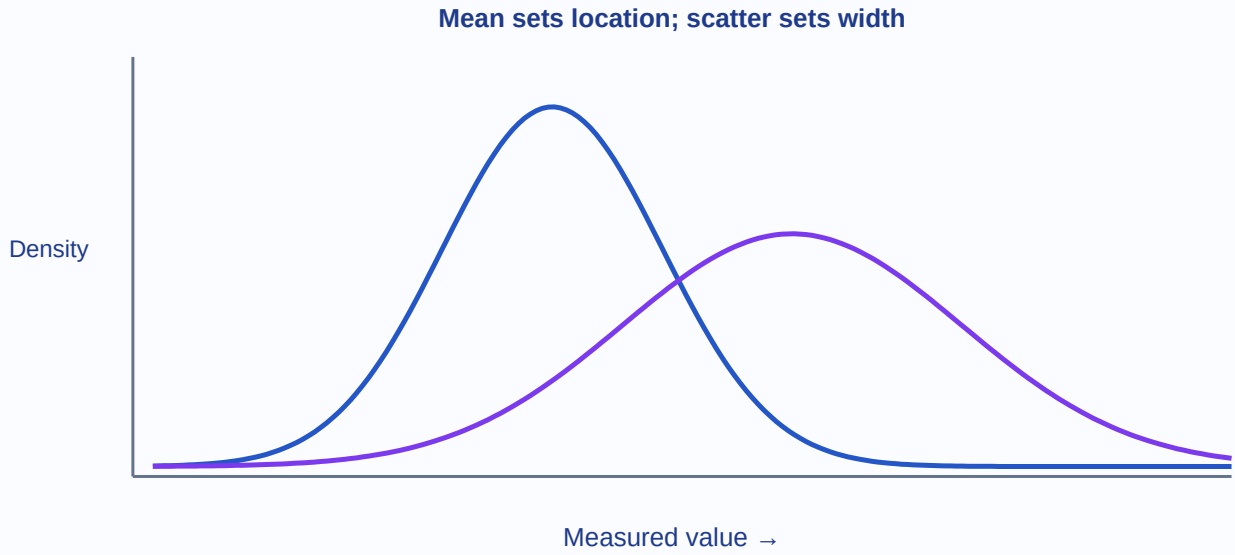


Figure 1. Normalized density curves: a narrower distribution is taller. Mean, median and standard deviation answer different questions.

2. Mean, median, spread and standard error

For n readings x_i,

$$\bar{x} = \frac{1}{n}\sum x_i, \qquad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}, \qquad \sigma_{\bar{x}} = \frac{s}{\sqrt{n}}. \tag{1}$$

The mean is sensitive to outliers; the median is robust. The standard deviation describes the scatter of individual readings. The standard error describes how well the mean is determined. Increasing n reduces the standard error as $1/\sqrt{n}$, but it does not remove a systematic offset.

3. Propagation of uncertainty

For independent random errors,

$$\sigma_f^2 \approx \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2. \tag{2}$$

For sums and differences, absolute uncertainties add in quadrature. For products and powers, fractional uncertainties add in quadrature with the appropriate powers. If $f = x^a y^b$, then

$$\left(\frac{\sigma_f}{f}\right)^2 \approx a^2 \left(\frac{\sigma_x}{x}\right)^2 + b^2 \left(\frac{\sigma_y}{y}\right)^2. \tag{3}$$

Known systematic offsets must be corrected, not averaged away. Independent, zero-mean standard uncertainties (including the residual uncertainty of a calibration) may be combined in quadrature; correlated terms require covariance or a separate systematic budget.

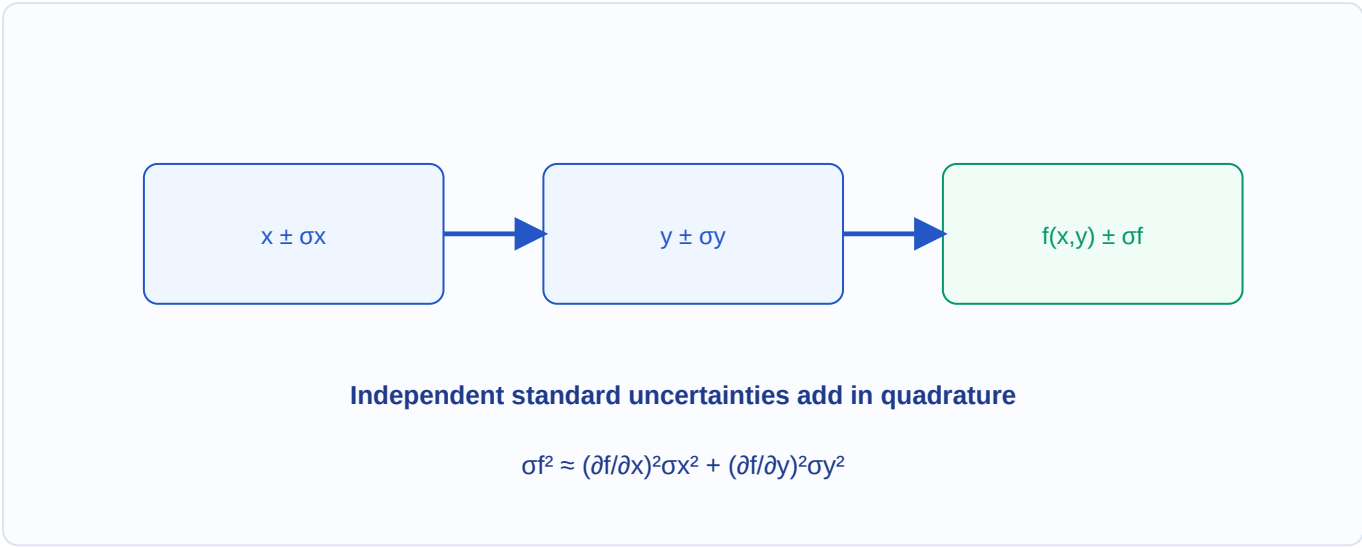


Figure 2. Known systematic offsets are corrected; correlated terms need separate treatment.

4. Distributions and confidence

Many measurement errors are approximately Gaussian. For a normal distribution, about 68% of values lie within one standard deviation, 95% within two, and 99.7% within three. These are useful rules of thumb, not guarantees. Small samples, outliers and non-Gaussian noise require caution.

A confidence interval describes the reliability of an estimation procedure, not the probability that a fixed true value is “random.” In Olympiad work, it is usually enough to state the uncertainty and the assumptions behind it.

5. Weighted means and outliers

If measurements have different uncertainties, weight them by inverse variance:

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}, \qquad w_i = \frac{1}{\sigma_i^2}, \qquad \sigma_{\bar{x}_w} = \frac{1}{\sqrt{\sum w_i}}. \tag{4}$$

Outliers should be investigated, not silently deleted. A bad point may reveal a cloud, a cosmic ray, a misidentified star or a transcription error. Robust statistics such as the median or a clipped mean can help, but the clipping rule must be stated.

6. Build an uncertainty budget

List every important source: photon noise, sky background, read noise, calibration, atmospheric extinction, timing, ruler reading, model assumptions. Correct known offsets, then combine independent, zero-mean standard uncertainties in quadrature; keep correlated systematics separate. Identify the dominant term. If one term dominates, improving the others will not help much.

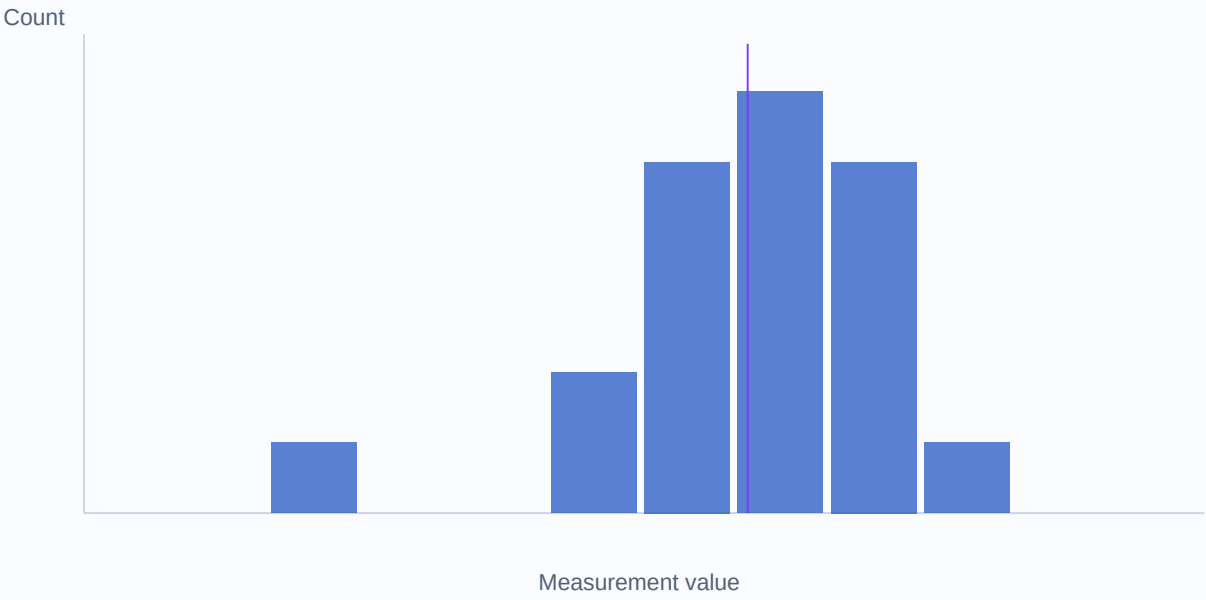
INTERACTIVE EXPERIMENT

Mean, median, scatter and the effect of one bad point.

A fixed synthetic sample is drawn from a normal-like distribution. The last point can be shifted into an outlier. Compare robust and non-robust summaries.

Mean	Median	Sample std dev
10.100	**10.121**	**0.294**

Standard error	Largest deviation
0.066	**0.904**



The mean is sensitive to outliers; the median is robust. Standard error describes uncertainty in the mean, not the spread of individual readings.

Try this. Add a large outlier. Which summary moves more? Does increasing n remove a systematic offset?

Snapshot of the experiment. Open the HTML edition to change the controls.

A reliable method. Separate random scatter from systematic bias. State the estimator, propagate uncertainties, check outliers, and report the dominant error source.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Mean and standard error. Five readings are 10.1, 9.9, 10.2, 10.0, 9.8. Find the mean, sample standard deviation and standard error.

Worked solution →

[1] Problem 2 — Median versus mean. A data set is 2, 3, 3, 4, 100. Find the mean and median. Which is more representative and why?

Worked solution →

[2] Problem 3 — Sum of uncertainties. Two independent lengths are 10.0 ± 0.2 cm and 5.0 ± 0.1 cm. Find their sum and uncertainty.

Worked solution →

[2] Problem 4 — Product uncertainty. A rectangle has sides 10.0 ± 0.1 cm and 5.0 ± 0.1 cm. Find the area and its uncertainty.

Worked solution →

[2] Problem 5 — Power law. If $y = x^3$ and $x = 2.00 \pm 0.02$, find y and its uncertainty.

Worked solution →

[2] Problem 6 — Weighted mean. Two measurements are 10.0 ± 0.2 and 10.5 ± 0.5. Find the weighted mean and its uncertainty.

Worked solution →

[1] Problem 7 — More readings. A single reading has standard deviation 0.3. How many independent readings are needed for the standard error of the mean to be 0.05?

Worked solution →

[2] Problem 8 — Systematic offset. A ruler reads 0.2 cm too long. Ten readings are averaged. Does the offset shrink? What is the correct way to handle it?

Worked solution →

[3] Problem 9 — Magnitude uncertainty. A flux has fractional uncertainty 5%. Estimate the corresponding magnitude uncertainty using $\sigma_m \approx 1.086\,\sigma_F/F$.

Worked solution →

[3] Problem 10 — Error budget. A photometric measurement has random uncertainty 0.03 mag, calibration uncertainty 0.04 mag and extinction uncertainty 0.02 mag. Estimate the combined standard uncertainty if the terms are independent and zero-mean. Which term dominates?

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.